

Hands-On-Learning: Building a Shared Responsibility Matrix in the Cloud

Scenario

You are a junior cloud security engineer at an organization beginning its migration to AWS. Leadership is confused about which security tasks belong to the **cloud provider** (AWS) versus which belong to the **customer**. To clarify, your task is to build a **Shared Responsibility Matrix** for different cloud service models (IaaS, PaaS, SaaS). This matrix will serve as a training artifact for engineers and a communication tool for executives.

By completing this lab, you will create a **visual matrix** that maps responsibilities across service models, improving organizational understanding of cloud security.

Instructions for Learners

1 Preparation

Make sure you have access to a drawing or documentation tool such as:

Lucidchart, Draw.io (diagrams.net), Miro, PowerPoint, or even Excel.

Review the three main cloud service models:

IaaS (Infrastructure as a Service)

PaaS (Platform as a Service)

SaaS (Software as a Service)

2 Designing Your Matrix Framework

Create a table with the following rows (security tasks):

Physical security

Network controls

Operating system patching

Identity & Access Management (IAM)

Data encryption (at rest & in transit)

Application security

Logging & monitoring

Add **columns** for: Provider vs. Customer responsibilities.

3 Populating Responsibilities

For IaaS (e.g., EC2):

Provider: Physical infra, hypervisor security

Customer: OS patching, IAM, encryption choices

For PaaS (e.g., RDS):

Provider: OS + database engine patching, backups

Customer: Data encryption, IAM, schema design

For SaaS (e.g., Google Workspace):

- Provider: Full infrastructure + app-level security
- Customer: Access controls, data usage, identity configuration

Clearly mark each responsibility in your table (e.g., ✓ Provider / ✓ Customer).

4 Creating a Visual Artifact

Format the matrix into a clear visual or infographic.

Use **colors or icons** to highlight:

- Shared tasks (both Provider + Customer)

Customer-only responsibilities

Save your matrix as a PDF, slide, or image for sharing.

5 Peer Review

Exchange your matrix with a peer or classmate.

Ask:

Did they classify responsibilities the same way?

Did they miss any tasks (e.g., compliance evidence, backups)?

Update your artifact based on feedback.

Reflection

Which responsibilities surprised you most (things you thought the provider handled but are actually customer tasks)?

How would this matrix change for a multi-cloud environment?

How could this matrix reduce risk in real-world onboarding and training?

Final Outcome

Created a **Shared Responsibility Matrix** for IaaS, PaaS, and SaaS.

Produced a peer-reviewed artifact suitable for training and executive communication.

Built a reusable tool to guide future cloud migrations and compliance efforts.

Next Steps

Expand the matrix to include **specific services** (EC2, RDS, S3, Google Workspace, etc.).

Layer in **framework mapping** (e.g., CIS, ISO, NIST) to show compliance overlap.

Use this matrix as a foundation for a **Cloud Security Onboarding Guide** in your organization.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.



Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.